

Homework 8

Monday, April 6, 2020 4:19 PM

(1) FT-VIII

1. $CO + * \rightleftharpoons CO^*$
2. $H_2 + 2* \rightleftharpoons 2H^*$
3. $CO^* + 2H \rightleftharpoons HCOH^* + 2*$
4. $HCOH^* + * \rightleftharpoons CH^* + OH^*$
5. $CH^* + H^* \rightleftharpoons CH_2^* + *$
6. $OH^* + H^* \rightleftharpoons H_2O + 2*$

$$\begin{aligned} r_1 &= -k_1 (C_{CO} - C_{CO^*}) \\ r_2 &= -k_2 (C_{H_2} - C_{H^*}) \\ r_3 &= -k_3 (C_{CO^*} \cdot [C_{H^*}]^2 - C_{HCOH^*}) \\ r_4 &= -k_4 (C_{HCOH^*} - C_{CH^*} \cdot C_{OH^*}) \\ r_5 &= -k_5 (C_{CH^*} \cdot C_{H^*} - C_{CH_2^*}) \\ r_6 &= -k_6 (C_{OH^*} \cdot C_{H^*} - C_{H_2O}) \end{aligned}$$

$$\frac{r_1}{k_1} = \frac{r_2}{k_2} = \frac{r_3}{k_3} = \frac{r_4}{k_4} = \frac{r_5}{k_5} = \frac{r_6}{k_6} = 0 \rightarrow \begin{aligned} C_{CO^*} &= C_{CO} - (1) \\ C_{H_2} &= C_{H^*} - (2) \\ C_{HCOH^*} &= C_{CH^*} \cdot C_{OH^*} - (4) \\ C_{CH^*} \cdot H^* &= C_{CH_2^*} - (5) \\ C_{OH^*} C_{H^*} &= C_{H_2O} - (6) \\ C_{OH^*} &= \frac{C_{H_2O}}{C_{H^*}} \text{ (from 6)} \end{aligned}$$

$$r_3 = -k_3 (C_{CO^*} [C_{H^*}]^2 - C_{HCOH^*})$$

$$r_3 = -k_3 (C_0 C_{H_2} - C_{CH^*} \cdot C_{OH^*})$$

$$r = r_3 = -k_3 \left(C_0 C_{H_2} - \frac{C_{CH^*} C_{H_2O}}{C_{H^*}} \right)$$

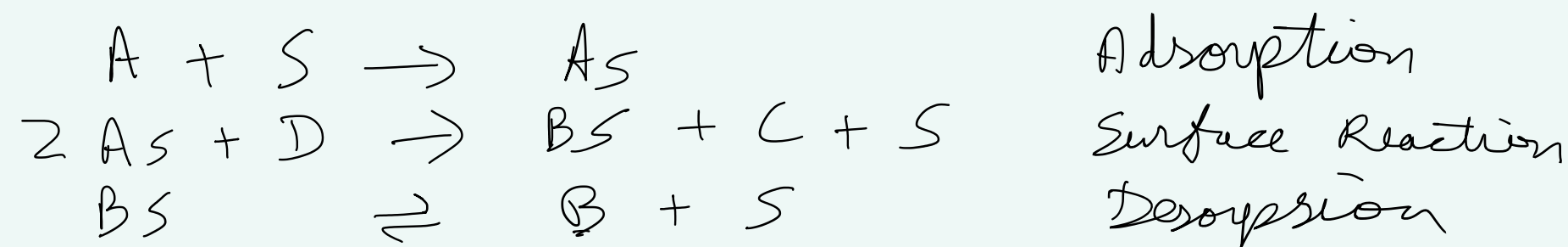
$$[CHCOH^*] = [CH^*] [OH^*]$$

$$r_3 = -k_3 \left([CO][H_2] - \frac{[CH^*][H_2O]}{[H^*]} \right)$$

overall rec: $CO + 2H_2 \rightleftharpoons CH_2^* + H_2O$

(2) $2A + D \rightarrow B + C$

$$r_A = \frac{-k P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2}$$



Surface reaction is rate limiting

$$r = k (C_{A_S}^2 P_D)$$

For Adsorption & Desorption,

$$C_{A_S} = K_A P_A C_v$$

$$C_{B_S} = K_B P_B C_v$$

Site Balance,

$$C_T = C_v + C_{A_S} + C_{B_S}$$

$$C_T = C_v + K_A P_A C_v + K_B P_B C_v$$

$$C_v = \frac{C_T}{1 + K_A P_A + K_B P_B}$$

$$r = k (C_{A_S}^2 P_D)$$

$$= k (K_A^2 P_A^2 C_v^2 P_D)$$

$$r = k \frac{K_A^2 P_A^2 P_D}{(1 + K_A P_A + K_B P_B)^2}$$